

A SERIES

Battery Connector for Industrial Vehicles

- 80 - 160 - 320 Sizes • UL and CSA Rated • RoHS Compliant
- Acid & Impact Resistant • Industrial Applications



APP
Anderson Power Products®

An **IDEAL** Company

A SERIES BATTERY CONNECTOR

The Anderson™ family of A Series Battery Connectors are expertly designed for ease of use and incorporate an innovative contact carrier to simplify assembly. The A Series connectors include standard 80, 160, and 320 sizes. Robust design and materials offer superior performance. All Materials chosen are RoHS compliant and selected to ensure years of reliability in adverse industrial environments. These attributes make the A Series connectors the ideal choice for a wide range of industrial applications in Material Handling, Battery Charging, Utility Vehicles, Motive Power, and Sweepers/Scrubbers amongst many others.



KEY ATTRIBUTES

Safety Agency Approvals - UL and CSA.



Voltage Keying - A series connectors offer keys for Dry-Cell, Wet-Cell, and Universal applications coded for 24/36/48/72/80 and 96 volts DC applications.

Sturdy Advanced Polymer Housings - Provide superior resistance to impact, acid and temperature extremes.

Silver Plated Copper Contacts - Provide excellent conductivity and superior mating cycle performance.

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SPECIFICATIONS

ELECTRICAL

Current Ratings (Amperes)

Wire Size:	A80		A160		A320	
	CSA*	UL**	CSA*	UL**	CSA*	UL**
10 mm ²	70	90	N/A	N/A	N/A	N/A
16 mm ²	90	120	90	120	N/A	N/A
25 mm ²	110	140	120	160	N/A	N/A
35 mm ²	120	160	140	180	N/A	N/A
50 mm ²	N/A	N/A	160	220	165	220
1/0 AWG	N/A	N/A	N/A	N/A	190	270
70 mm ²	N/A	N/A	N/A	N/A	200	260
3/0 AWG	N/A	N/A	N/A	N/A	240	300
95 mm ²	N/A	N/A	N/A	N/A	245	425
4/0 AWG	N/A	N/A	N/A	N/A	250	425
120 mm ²	N/A	N/A	N/A	N/A	260	450

* CSA allows for maximum temperature rise = 30°C

** UL allows for total temperature (based on min. 25°C ambient) = 105°C

Voltage Ratings

	A80		A160		A320	
	CSA	UL	CSA	UL	CSA	UL
AC / DC	600V	600V	600V	600V	600V	600V
DWV (AC)	2200		2200		2200	

Wire Range

	A80		A160		A 20	
	mm ²	AWG	mm ²	AWG	mm ²	AWG
Power - Min.	10	#8	16	#6	50	1/0
Power - Max.	35	#2	50	1/0	120	4/0
Upper Aux. - Min.	1.5	#18	4	#12	4	#12
Upper Aux. - Max.	2.5	#14	4	#12	4	#12
Lower Pilot Aux. - Min.	1.5	#18	6	#10	6	#10
Lower Pilot Aux. - Max.	2.5	#14	6	#10	6	#10

Average Initial Contact Resistance Across Mated Connector- micro-ohms

	A80	A160	A320
Avg. Contact Resistance - Power	65	55	30
Avg. Contact Resistance - Aux.	1100	920	1000

Operating Temperature (all Series): -25 to 75°C

MECHANICAL

Misc. Data

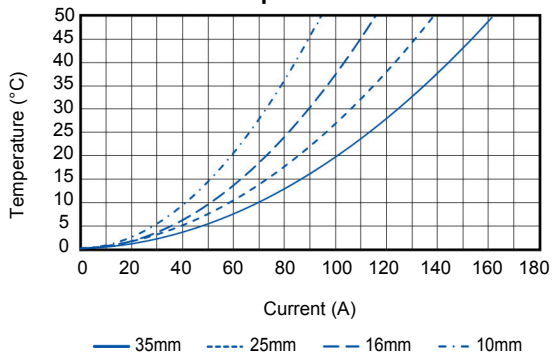
	A80	A160	A320
Avg. Mating Force N (lb)	68 (15.3)	170 (38.2)	71 (16.0)
Contact Retention N (lb)	134 (30) min	445 (100) min	445 (100) min
Degree of Protection	IP23	IP23	IP23
Flammability	UL94 HB, Horizontal Burn		

MATERIALS

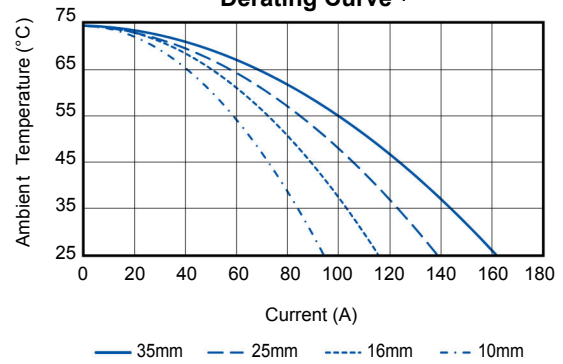
	A80 - A160 - A320
Housings	PA6 (Nylon) glass filled
Contacts	Copper alloy silver plate
Contact Plating - Power (min.)	Silver Plating - 6 microns
Contact Plating - Auxiliary (min.)	Silver Plating - 6 microns
Hardware	Steel, zinc chromate plate

TEMPERATURE CHARTS

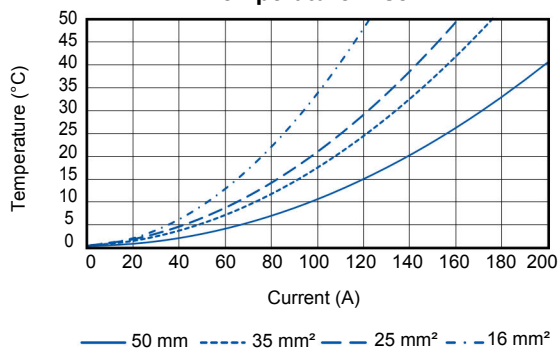
**A80
Temperature Rise**



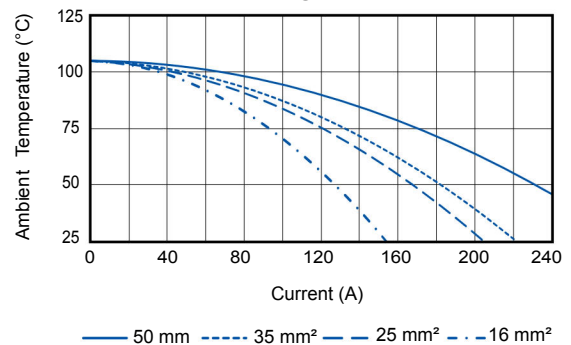
**A80
Derating Curve ¹**



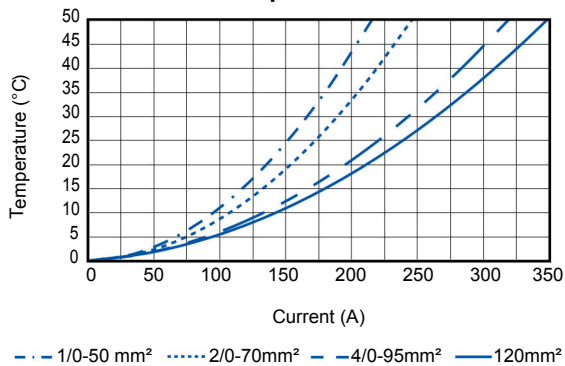
**A160
Temperature Rise**



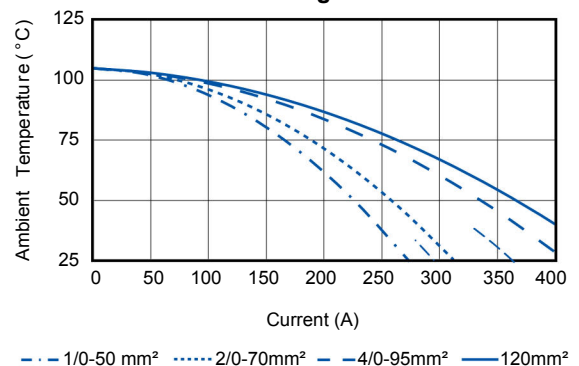
**A160
Derating Curve ¹**



**A320
Temperature Rise**



**A320
Derating Curve ¹**

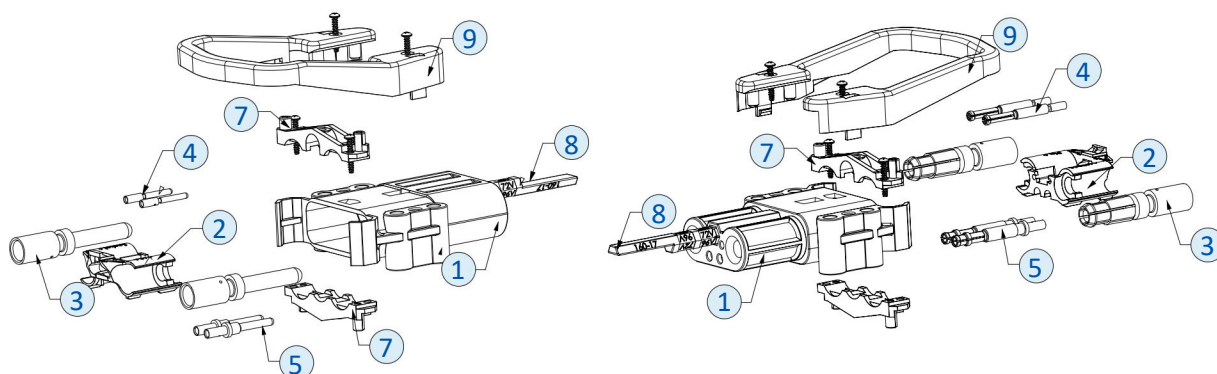


¹ 0.8 Derating per IEC 60512-3

NOTE 1: Temperature ratings based on power contacts energized only.

NOTE 2: Temperature rise charts are based on a 25°C ambient temperature.

ORDERING INFORMATION



Components

Item Number	Description - mm ² (AWG)	A80		A160		A320	
		Male	Female	Male	Female	Male	Female
3	Power Contacts						
	Power Contact 10 mm ² (8 AWG)	80-1010	80-1110	N/A	N/A	N/A	N/A
	Power Contact 16 mm ² (6 AWG)	80-1016	80-1116	160-1016	160-1116	N/A	N/A
	Power Contact 25 mm ² (4 AWG)	80-1025	80-1125	160-1025	160-1125	N/A	N/A
	Power Contact 35 mm ² (2 AWG)	80-1035	80-1135	160-1035	160-1135	N/A	N/A
	Power Contact 50 mm ² (1 AWG to 1/0)	N/A	N/A	160-1050	160-1150	320-1050	320-1150
	Power Contact 70 mm ² (2/0 to 3/0)	N/A	N/A	N/A	N/A	320-1070	320-1170
	Power Contact 95 mm ² (3/0 to 4/0)	N/A	N/A	N/A	N/A	320-1095	320-1195
	Power Contact 120 mm ² (4/0)	N/A	N/A	N/A	N/A	320-1004	320-1104
4	Upper Auxiliary Contact 4mm ² (12 AWG)	NA	NA	160-14	160-15	160-14	160-15
	Upper Auxiliary Contact 1.5mm/2.5 mm - (18/14 AWG)	E80-32	E80-33	320-24	320-25	320-24	320-25
5	Lower Pilot Auxiliary Contact 6mm ² (10 AWG)	NA	NA	160-12	160-13	160-12	160-13
	Lower Pilot Auxiliary Contact 1.5mm / 2.5 mm - (18/14 AWG)	E80-30	E80-31	320-22	320-23	320-22	320-23
8	Voltage Coding Pins						
	Grey Coding Pin - Wet (160 and 320 use same Coding Pins)	80-9	80-9	160-9	160-9	160-9	160-9
	Green Coding Pin - Dry (160 and 320 use same Coding Pins)	80-17	80-17	160-17	160-17	160-17	160-17
	* Yellow Coding Pin - Universal (160 and 320 use same Coding Pins)	80-19	Not Used	160-19	Not Used	160-19	Not Used
9	** Handle - High Profile with screws	N/A	N/A	32-89	32-89	32-89	32-89
	** Handle - Low Profile with screws	80-89	80-89	16-89	16-89	16-89	16-89
	Handle Red - Low Profile with screws	80-8-R	80-8-R	N/A	N/A	N/A	N/A
1	Housing Only	80-1	80-2	160-1	160-2	320-1	320-2
2	Contact Holder	80-5B	80-5B	160-5	160-5	320-5	320-5
7	Cable Clamps with Screws	80-67	80-67	160-67	160-67	320-67	320-67
1+2+7+8	Male Housing with Contact Holder and Cable Clamp with Screws (Gray Key) - Individual Pack	A80400-0009		A16400-0009		A32400-0009	
1+2+7+8	Female Housing with Contact Holder, **** Lower Pilot Auxiliary Holder, Cable Clamps with Screws - (Gray Key) - Individual Pack		A80500-0009		A16500-0009		A32500-0009
1+2+7+8	Male Housing with Contact Holder and Cable Clamp with Screws (Green Key) - Individual Pack	A80400-0209		A16400-0209		A32400-0209	
1+2+7+8	Female Housing with Contact Holder, **** Lower Pilot Auxiliary Holder, Cable Clamps with Screws - (Green Key) - Individual Pack		A80500-0209		A16500-0209		A32500-0209
1+2+7+8	Male Housing with Contact Holder and Cable Clamp with Screws (Yellow Key) - Individual Pack	A80400-0309		A16403-0309		A32400-0309	

* Universal yellow coding key is only used on male connectors

** ABC 160 and ABC 320 handles work with both the ABC 160 and ABC 320 series connectors

PART NUMBER CONFIGURATOR

A80 Connector Part Number Selection

Series	Gender	Main Contact	Handle	Coding Key	Auxiliary Contacts	Packing
A80	4 or 5	25	- 1	0	0	9
4	Male					9 Individual
5	Female					8 Bulk
00	None - Order Separately				0 None	
10	10 mm ² (#8 AWG)				A (2) Lower Pilot Auxiliary Contacts	
16	16 mm ² (#6 AWG)				B (2) Upper Auxiliary Contacts	
25	25 mm ² (#4 AWG)				C (2) Lower Pilot Auxiliary Contacts & (2) Upper Auxiliary Contacts	
35	35 mm ² (#2 AWG)					
0	No		0	Grey, Wet Cell		
1	Black		2	Green, Dry Cell		
2	Red		3	Yellow, Universal (use with male housing only)		

A160 Connector Part Number Selection

Series	Gender	Main Contact	Handle	Coding Key	Auxiliary Contacts	Packing
A16	4 or 5	25	- 1	0	0	9
4	Male					9 Individual
5	Female					8 Bulk
00	None - Order Separately				0 None	
16	16 mm ² (#6 AWG)					mm ² AWG
25	25 mm ² (#4 AWG)				A (2) Lower Auxiliary Contacts	6 10
35	35 mm ² (#2 AWG)				B (2) Upper Auxiliary Contacts	4 12
50	50 mm ² (1/0 AWG to #1 AWG)				C (2) Lower Auxiliary Contacts & (2) Upper Auxiliary Contacts	6 10
0	No					4 12
1	Black				D (2) Lower Auxiliary Contacts	1.5/2.5 18/14
					E (2) Upper Auxiliary Contacts	1.5/2.5 18/14
					F (2) Lower Auxiliary Contacts & (2) Upper Auxiliary Contacts	1.5/2.5 18/14
			0	Grey, Wet Cell		1.5/2.5 18/14
			2	Green, Dry Cell		
			3	Yellow, Universal (use with male housing only)		

A320 Connector Part Number Selection

Series	Gender	Main Contact	Handle	Coding Key	Auxiliary Contacts	Packing
A32	4 or 5	50	- 1	0	0	9
4	Male					9 Individual
5	Female					8 Bulk
00	None - Order Separately				0 None	
50	50 mm ² (1/0 AWG to #1 AWG)					mm ² AWG
70	70 mm ² (2/0 AWG to 3/0 AWG)				A (2) Lower Auxiliary Contacts	1.5/2.5 18/14
95	95 mm ² (3/0 AWG to 4/0 AWG)				B (2) Upper Auxiliary Contacts	1.5/2.5 18/14
12	120 mm ² (4/0 AWG)				C (2) Lower Auxiliary Contacts & (2) Upper Auxiliary Contacts	1.5/2.5 18/14
0	No					1.5/2.5 18/14
1	Black				D (2) Lower Auxiliary Contacts	6 10
					E (2) Upper Auxiliary Contacts	4 12
					F (2) Lower Auxiliary Contacts & (2) Upper Auxiliary Contacts	6 10
			0	Grey, Wet Cell		4 12
			2	Green, Dry Cell		
			3	Yellow, Universal (use with male housing only)		

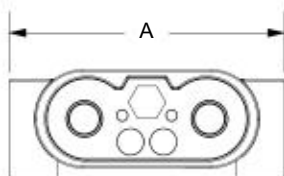
Packaging: Individual (Kits) - all components for a complete connector are packaged together in individual bags within one box. Bulk (Kits) - all components required are packaged separately by part number selected in one bag/box. **Handles:** A80 comes with the low profile T handle • A160 comes with the low profile handle • A320 comes with the high profile handle.

DRAWINGS

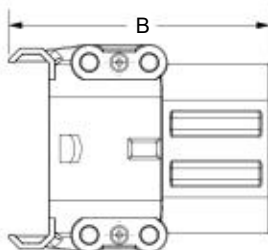


Housing Dimensions							
		A80		A160		A320	
		mm	in	mm	in	mm	in
Male Housings	A	68.0	2.68	83.5	3.29	89.5	3.52
	B	74.4	2.93	103.8	4.09	123.5	4.86
	C	26.1	1.03	33.4	1.31	40.8	1.61
Female Housings	A	68.0	2.68	83.5	3.29	89.5	3.52
	B	73.8	2.91	104.0	4.09	117.0	4.61
	C	26.1	1.03	32.6	1.28	37.6	1.48

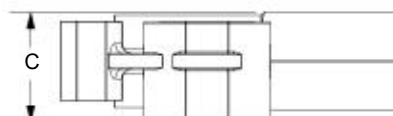
Male Front View



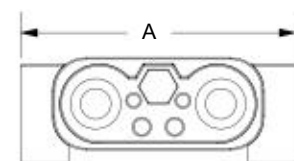
Male Top View



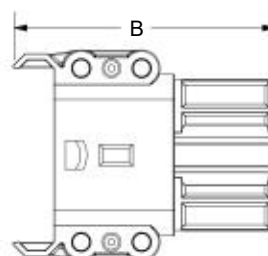
Male Side View



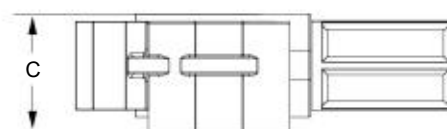
Female Front View



Female Top View



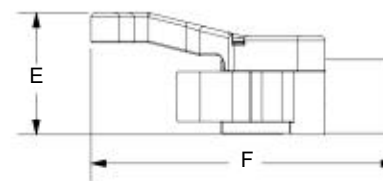
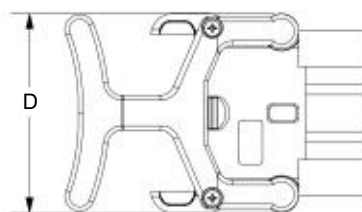
Female Side View



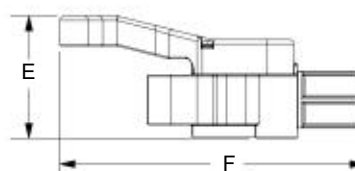
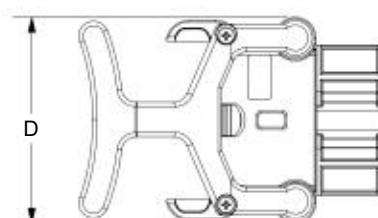
Dimensions of Housings With Handles

		A80	
		mm	in
Male Housings	D	68.0	2.68
	E	41.2	1.62
	F	110.6	4.35
Female Housings	D	70.0	2.76
	E	41.2	1.62
	F	110.6	4.35

A80 Male Housing with Handle

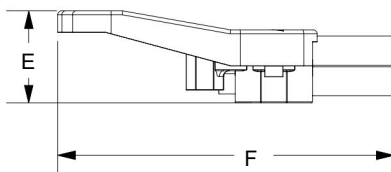
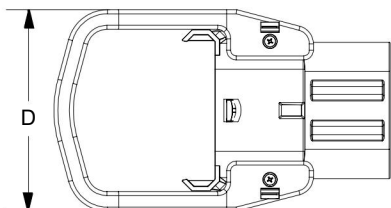


A80 Female Housing with Handle

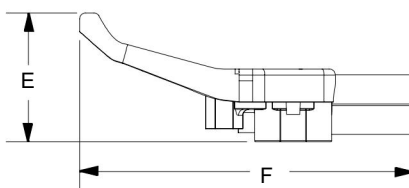
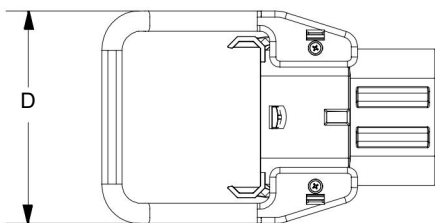


A160 Handles											
		Low Profile		High Profile				Low Profile		High Profile	
		mm	in	mm	in			mm	in	mm	in
Male Housings	D	101	4.0	106	4.2	Female Housings	D	101	4.0	106	4.2
	E	46	1.8	64	2.5		E	46	1.8	64	2.5
	F	168	6.6	167	6.6		F	173	6.8	172	6.8

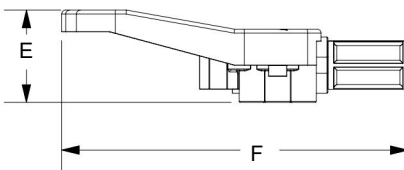
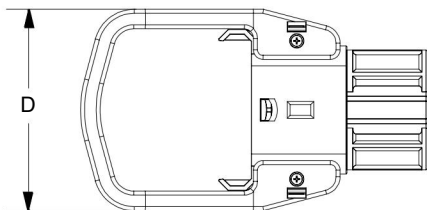
Male Housing Low Profile Handle



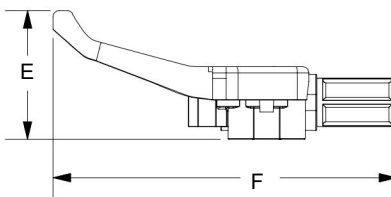
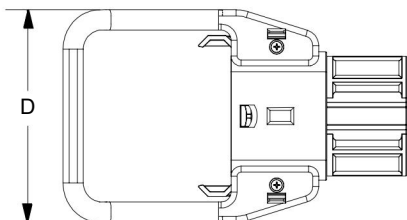
Male Housing High Profile Handle



Female Housing Low Profile Handle



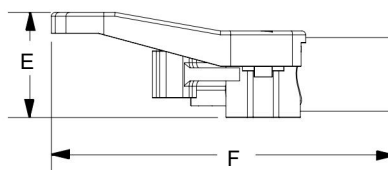
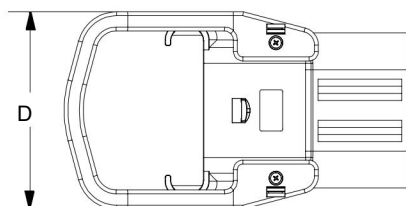
Female Housing High Profile Handle



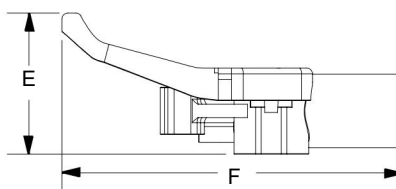
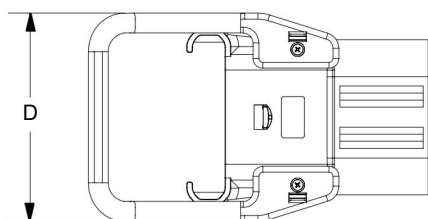
A320 Handles

		Low Profile		High Profile				Low Profile		High Profile	
		mm	in	mm	in			mm	in	mm	in
Male Housings	D	101	4.0	106	4.2	Female Housings	D	101	4.0	106	4.2
	E	54	2.1	72	2.8		E	53	2.1	71	2.8
	F	175	6.9	174	6.9		F	171	6.7	173	6.8

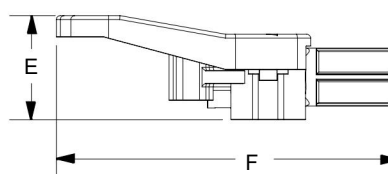
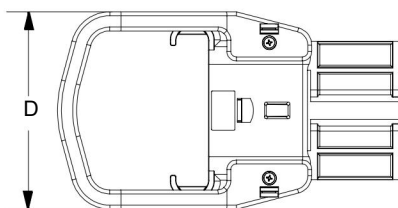
Male Housing Low Profile Handle



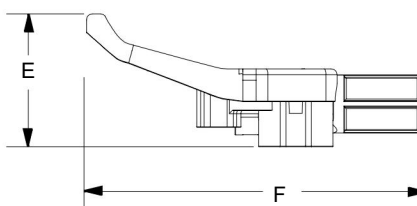
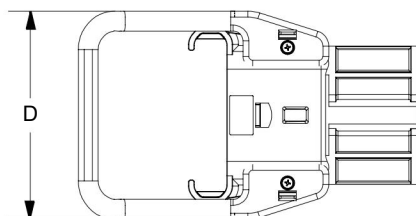
Male Housing High Profile Handle



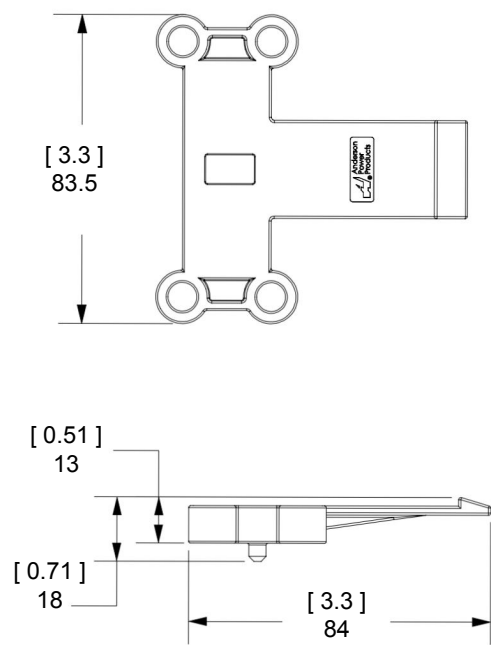
Female Housing Low Profile Handle



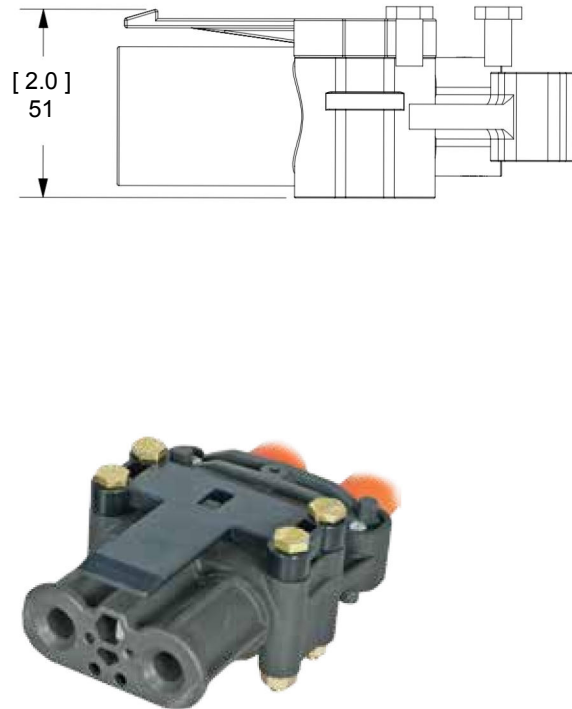
Female Housing High Profile Handle



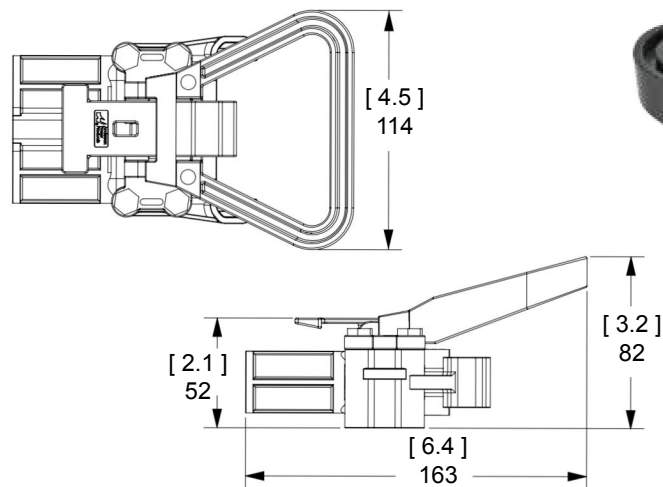
Latch Plate 160 & 320
P/N: A320-LP



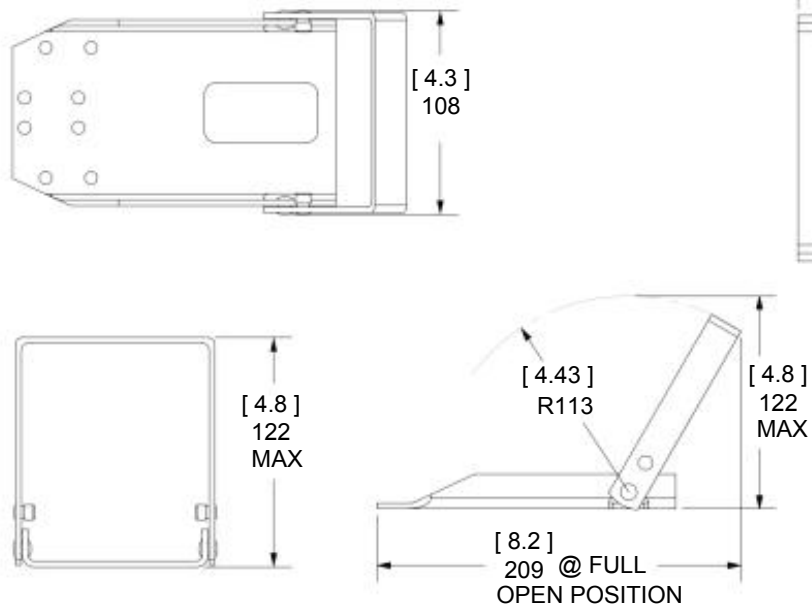
Latch Plate on
A160 & 320 Male Housing



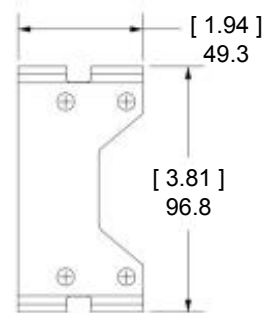
Handle & Lever Assembly
on A160 & 320 Female Housing:
P/N: A320HL



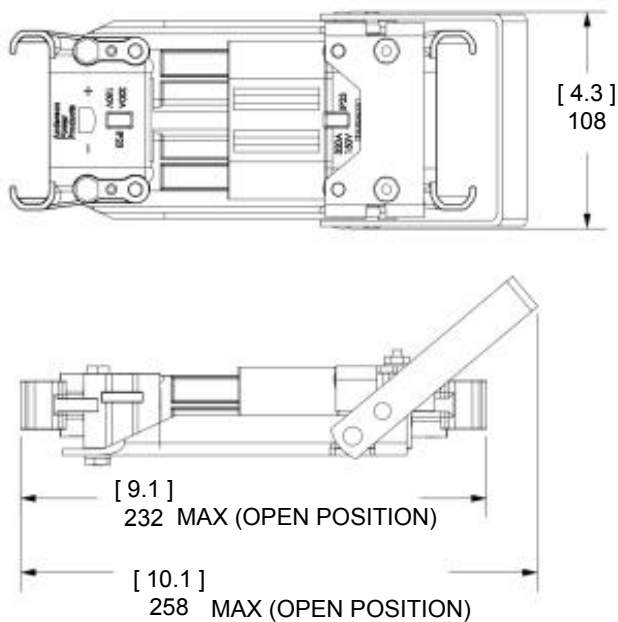
Manual Release for A160 & 320 P/N: 994G4



Battery Bracket - P/N: 993G4



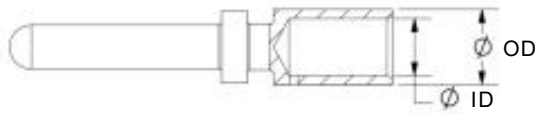
Manual Release with Connectors



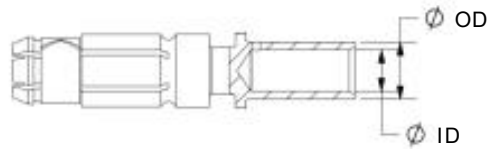
Contacts						
Series	Wire Size		OD		ID	
	Metric	AWG	mm	in	mm	in
A80	10 mm ²	#6 to 8	6.0	0.24	4.5	0.18
A80 / 160	16 mm ²	#4 to 6	8.4	0.33	6.0	0.24
	25 mm ²	#2 to 4	11.0	0.43	8.0	0.31
	35 mm ²	#1 to 2	12.5	0.49	9.0	0.35
A160 / 320	50 mm ²	#1 to 1/0	14.5	0.57	11.0	0.43
A320	70 mm ²	2/0 to 3/0	17.0	0.67	13.0	0.51
	95 mm ²	3/0 to 4/0	19.8	0.78	15.0	0.59
	95 mm ²	3/0 to 4/0	19.8	0.78	15.0	0.59
	120 mm ²	4/0	19.8	0.78	15.6	0.61



Male Power Contact



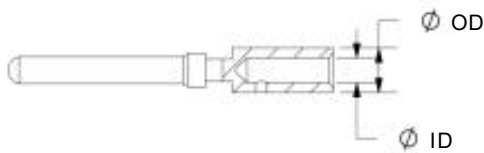
Female Power Contact



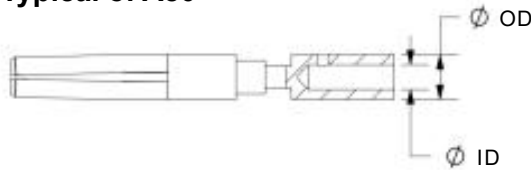
Auxiliary Contacts

Series	Part Number	Position	mm ²	AWG	OD		ID	
					mm	IN	mm	IN
A80	E80-32 / E80-33	Upper	1.5 / 2.5	18 / 14	3.9	0.15	2.2	0.09
	E80-30 / E80-31	Lower	1.5 / 2.5	18 / 14	3.9	0.15	2.2	0.09
A160	160-14 / 160-15	Upper	4	12	4.1	0.16	2.9	0.11
	320-24 / 320-25	Upper	1.5 / 2.5	18 / 14	4.6	0.18	2.2	0.09
A320	160-12 / 160-13	Lower	6	10	5	0.2	3.9	0.15
	320-22 / 320-23	Lower	1.5 / 2.5	18 / 14	4.6	0.18	2.2	0.09

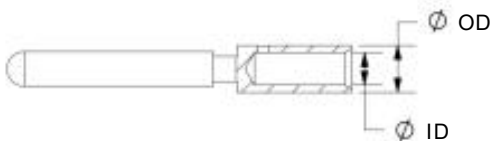
Upper Auxiliary Pin Typical of A80



Upper Auxiliary Socket Typical of A80



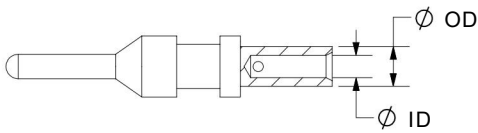
Upper Auxiliary Pin Typical of A160 / 320



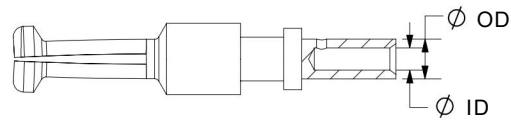
Upper Auxiliary Socket Typical of A 160/ 320



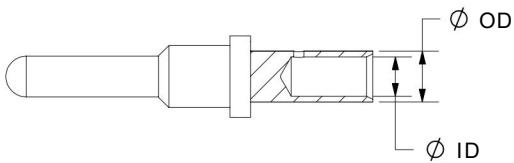
Lower Pilot Auxiliary Pin Typical of A80



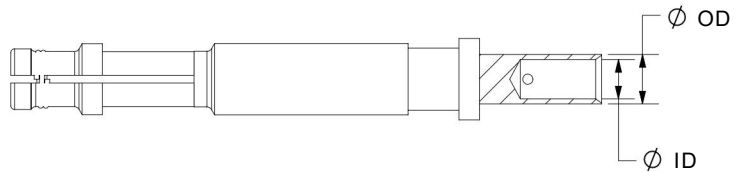
Lower Auxiliary Socket Typical of A80



Lower Pilot Auxiliary Pin Typical of A160 / 320

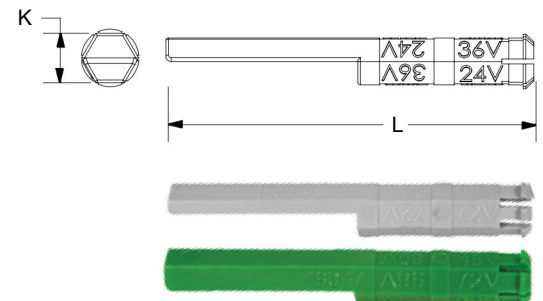


Lower Auxiliary Socket Typical of A160 / 320



Wet / Dry Voltage Key					Universal Voltage Key			
Series	K		L		K		L	
	mm	in	mm	in	mm	in	mm	in
A80	8.0	0.31	46	1.77	8.0	0.31	46	1.77
A160 / 320	9.3	0.37	69	2.72	9.3	0.37	69	2.72

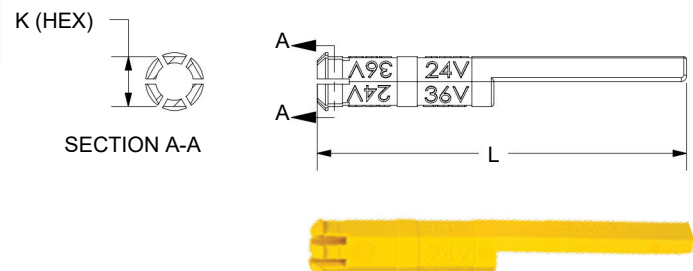
Wet / Dry Voltage Key - Grey or Green



Keying Plug	Identification	Assembled In
	Used for Wet Cell Battery /Charger – Color Grey	Battery Receptacle and Charging Plug
	Used for Dry Cell Battery/Charger – Color Green	Battery Receptacle and Charging Plug
	Used for Vehicle only - Yellow (Universal)	Vehicle Plug only - used when vehicle may use wet or dry cell batteries.

Universal Voltage Key - Yellow (Use with male housing only)

Universal (Yellow) voltage keys are to be used exclusively at the vehicle side to permit both the use of wet-cell and dry-cell batteries.



TOOLING INFORMATION

A Series 80A, 160A, & 320A Hydraulic Tooling								
Wire Size		Part Numbers	Loose Piece Contact Crimp Tools					
mm ²	AWG	Power Contact	Hydraulic Hand Bench Tool	+	Die	OR	Hydraulic Hand Bench Tool	
10 mm ²	#8 to #6	80-1010	1387G3 OR 1370G1		1322G16		N/A	
		80-1110						
	#6	80-1016			1322G9	1368 Series		
		80-1116						
16 mm ²		80-1016			1322G10			
		80-1116						
25 mm ²	#4 to #2	80-1025			1322G15			
		80-1125						
35 mm ²	#2 to #1	80-1035	N/A	N/A				
		80-1135						
16 mm ²	#6 to #4	160-1016	1387G3 OR 1370G1	1322G9				
		160-1116						
25 mm ²	#4 to #2	160-1025		1322G12				
		160-1125						
35 mm ²	#2 to #1	160-1035		1322G7				
		160-1135						
50 mm ²	#1 to 1/0	160-1050		1322G4				
		160-1150						
50 mm ²	1/0 AWG	320-1050	1387G3 OR 1370G1	1322G4				
		320-1150						
70 mm ²	2/0 AWG	320-1070		1322G3 (K) 8* 1322G2				
		320-1170						
95 mm ²	3/0 AWG	320-1095		1322G1				
		320-1195						
120 mm ²	4/0 AWG	320-1004 320-1104	N/A	N/A				
Pilot/Auxiliary Contacts								
ALL		Pins	SOLDER ONLY					
		Sockets						

*Use 1322G3 with K stranded wire only.



1370G1 & 1387G3



1368



1368-NL



1368-B

A Series 80A, 160A, & 320A Pneumatic Tooling

Wire Size		Part Numbers	Loose Piece Contact Crimp Tools				
mm ²	AWG	Power Contact	Pneumatic Bench Tool	+	Die	+	Locator (Single Crimp)
10 mm ²	#8 to #6	80-1010	N/A		N/A		N/A
		80-1110					
16 mm ²	#6 to #4	80-1016					
		80-1116					
25 mm ²	#4 to #2	80-1025					
		80-1125					
35 mm ²	#2 to #1	80-1035					
		80-1135					
mm ²	AWG	Power Contact	Pneumatic Bench Tool	+	Die	+	Locator (Double Crimp)
16 mm ²	#6 to #4	160-1016	1387G2		1303G14		1304G21 SOCKET 1304G20 PIN
		160-1116					
25 mm ²	#4 to #2	160-1025					
		160-1125					
35 mm ²	#2 to #1	160-1035					
		160-1135					
50 mm ²	#1 to 1/0	160-1050					
		160-1150					
50 mm ²	#1 to 1/0	320-1050	1387G2		1303G8		1304G29 Pins 1304G18 Sockets
		320-1150					
70 mm ²	2/0 AWG	320-1070					
		320-1170					
95 mm ²	3/0 AWG	320-1095					
	4/0 AWG	320-1195					
120 mm ²	4/0 AWG	320-1004					
		320-1104					
Pilot/Auxiliary Contacts							
ALL		All Crimp Pins	SOLDER ONLY				
		All Crimp Sockets					



1387G1 & 1387G2

ADDITIONAL PRODUCTS FOR INDUSTRIAL VEHICLES

EBC Series

- Meets DIN 43589-1
- Meets EN1175-1
- Chemical Resistant Plastic
- RoHS Compliant



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